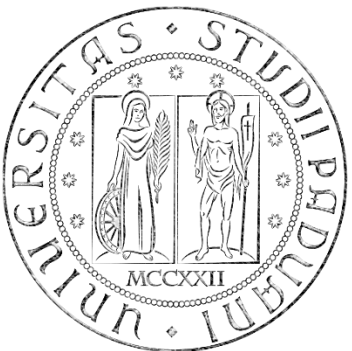


Biomedical Wearable Technologies
for Healthcare and Wellbeing

State Management

A.Y. 2024-2025

Giacomo Cappon



Outline

- **State management concepts**

- Provider

- Case study

- Exercise

- Homework

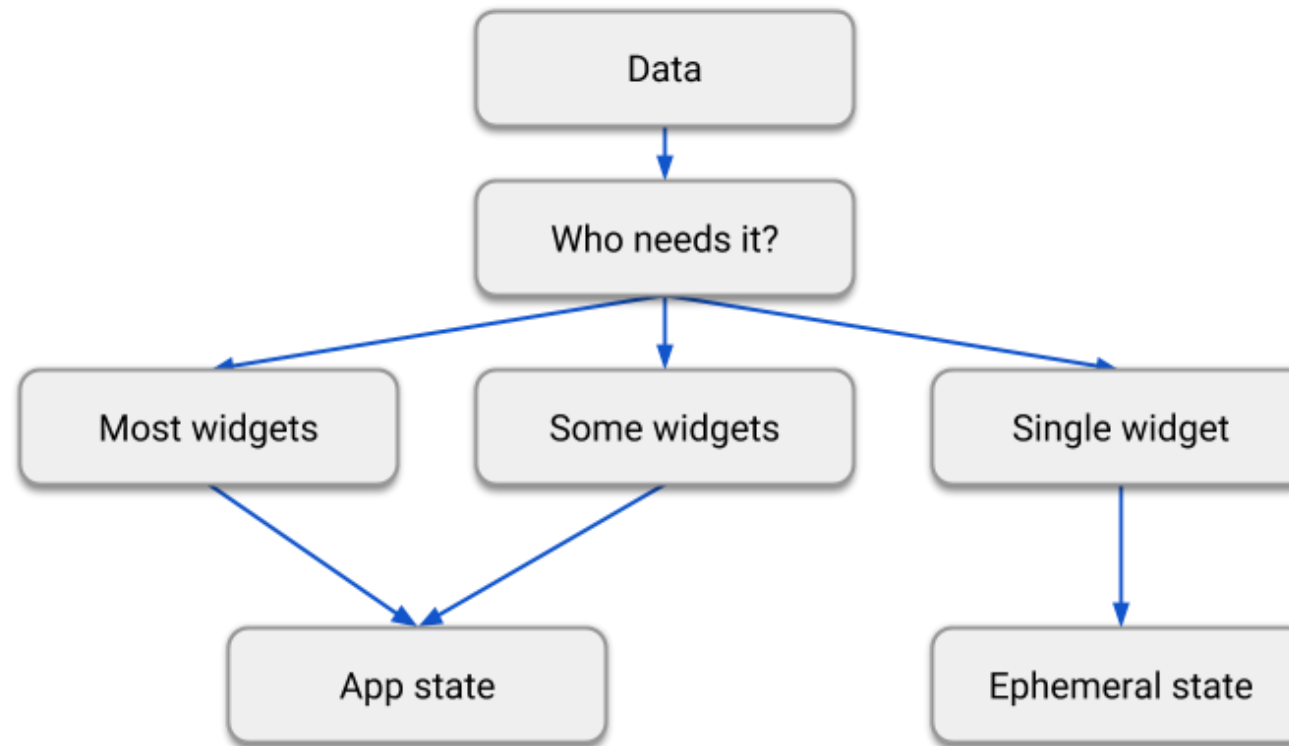
- Resources

State

- State stands for everything that is necessary to define how the app and its screen behave and look at some point in time:
 - Assets
 - Variables
 - Fonts
 - ...
- Conceptually can be divided in:
 - **Ephemeral state**: sometimes called *local state*, what can be strictly contained in a Widget
 - **App state**: sometimes called *shared state*, things that you want to share across many parts of the app

State

- There is no universal rule to choose if a variable is part of the ephemeral state or the app state. A diagram that can help:



Remember: Flutter is declarative

- Flutter is a declarative framework

$$\text{UI} = f(\text{state})$$

The layout
on the screen

Your
build
methods

The application state

- State is changed? Build methods are called and the UI is refreshed.

So far...

- So far, we had a grasp on state management:
 - Stateful widgets: their state can change through time
 - Stateless widgets: their state cannot change though time
- We need to understand how to change state when something occurs (e.g., a button is pressed, data change,...) and reflect those changes to the UI (rebuilding it).
- In lesson 4, we used `setState()`:

So far...

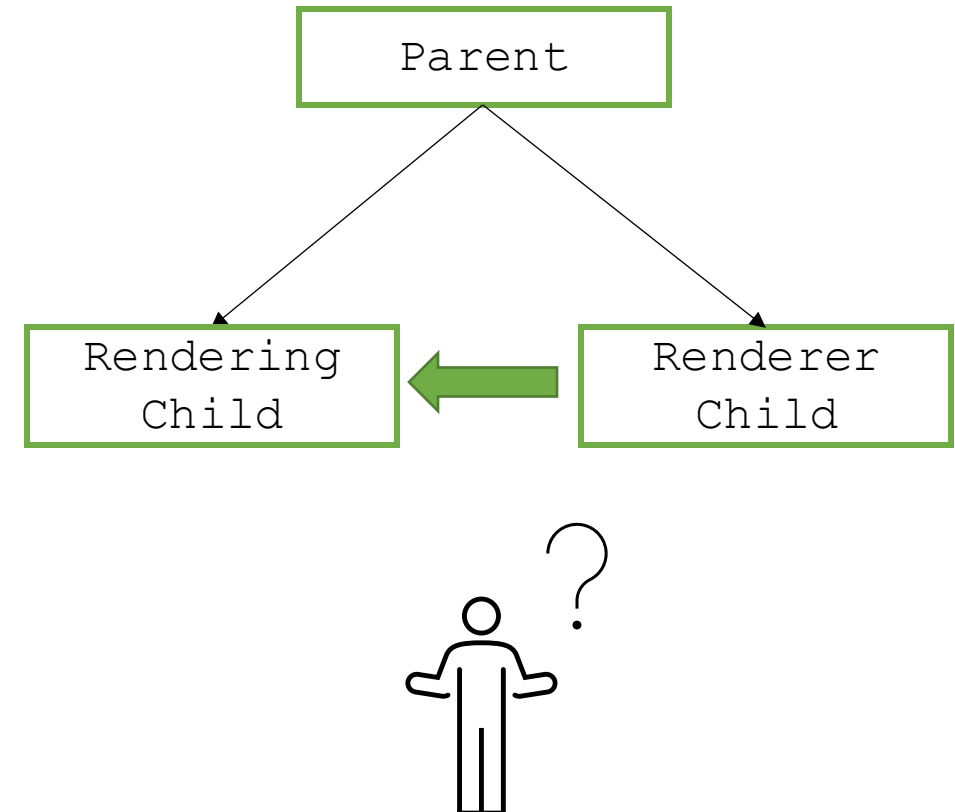
➤ In lesson 4, we used setState():

```
...
@override
Widget build(BuildContext buildContext){
  return Column(
    mainAxisAlignment: MainAxisAlignment.center,
    children: [
      Text('Hello, $_word!'),
      ElevatedButton(onPressed: _changeWord, child: const
Text('Press me')),
    ],);
} //build

void _changeRandomWord() {
  setState() {
    _word = WordPair.random().first;
  };
} // _changeRandomWord
...
```

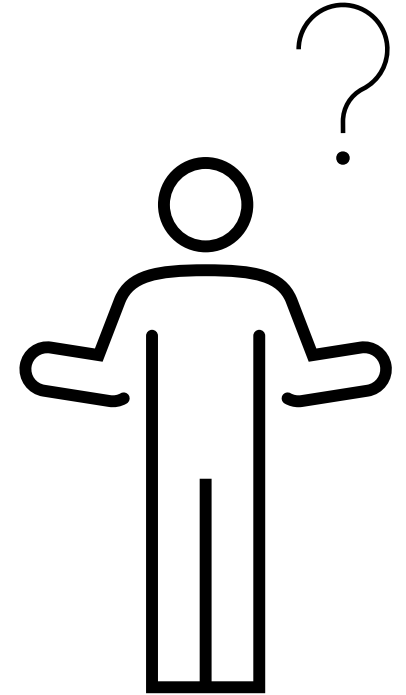
The limitation

- Why we cannot fully depend on this approach?
- In a typical widget tree, when two children of a parent are not in the same class as of the parent, rendering one child from another becomes impossible until and unless you involve the parent for the same.
- You either have to call a `setState()` from parent which in turn renders both children or pass a call-back function from one child to another child via the Parent.
- In other words, app state is very messy to handle



So, how to manage state?

- There is no such a thing as a “*universal way to manage state*”
- There are a lot of possible approaches:
 - Provider
 - Riverpod
 - Redux
 - BLoC
 - ...
- Every approach has its PROs and CONs. So?
- Here, we will discuss **Provider**, the recommended approach by the Flutter community.

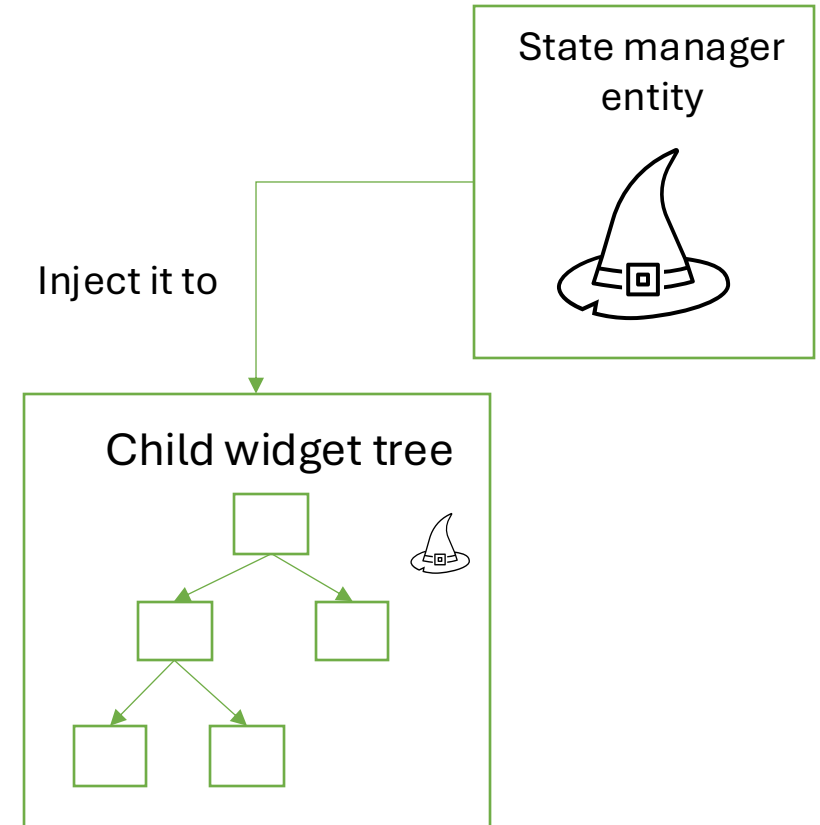


Outline

- State management concepts
- **Provider**
- Case study
- Exercise
- Homework
- Resources

Provider core idea

- Provider wants to provide!
- The core idea is to provide the entity (one or more classes) in charge of maintaining the state down through the widget tree
- Each child will be able to access to the entity and react to state changes.



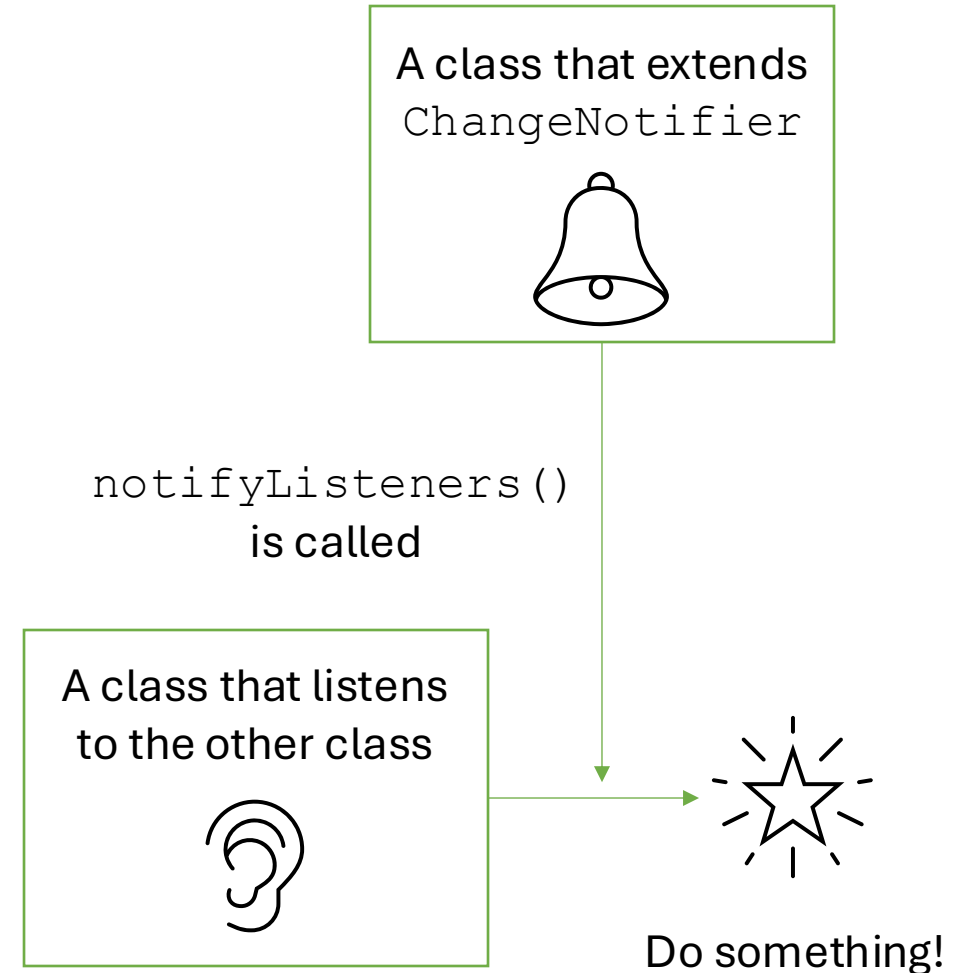
Provider classes

➤ Provider implement a set of classes, here's the most important:

- `ChangeNotifier`
- `ChangeNotifierProvider`
- `Consumer`
- `MultiProvider`
- ...

Provider classes: ChangeNotifier

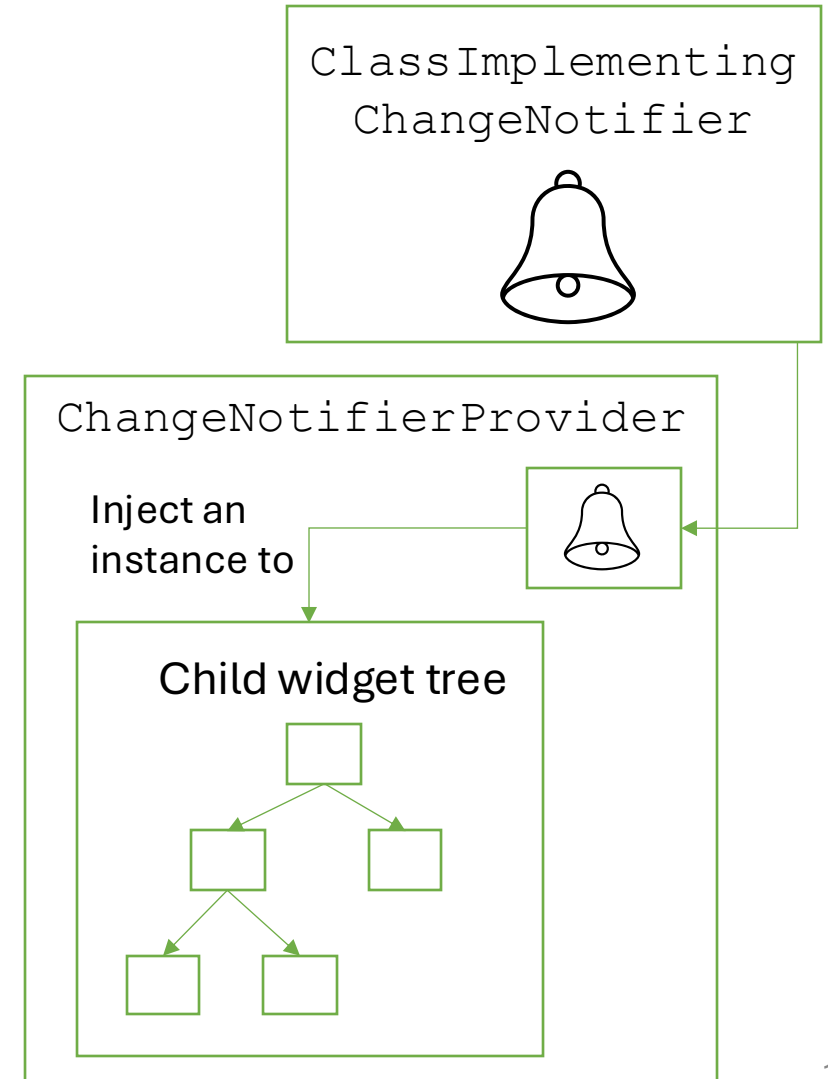
- `ChangeNotifier` is a class that can notify **listeners** of any changes in the state.
- You usually extend the `ChangeNotifier` for models so you can send notifications when your model changes.
- When something in the model changes, you call **`notifyListeners()`** and whoever is listening can use the newly changed model.



Provider classes: ChangeNotifierProvider

- `ChangeNotifierProvider` is a class that wraps a class that implements `ChangeNotifier` and **provide it** to its descendants.
- Now the widget tree can access to it (and use it!)
- Syntax:

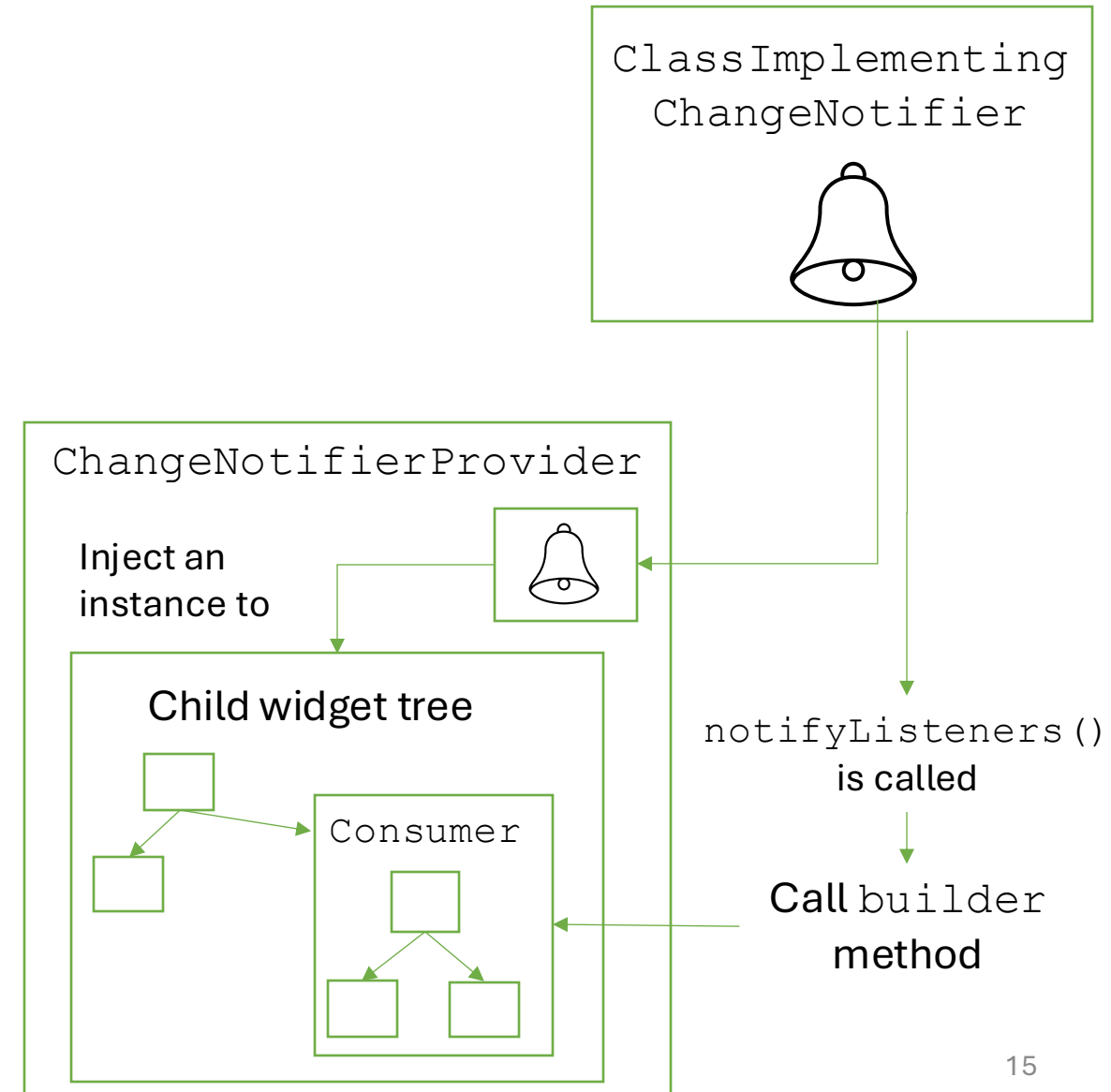
```
ChangeNotifierProvider(  
  create: (context) =>  
    ClassImplementingChangeNotifier(),  
  child: <widget_tree>,  
);
```



Provider classes: Consumer

- Consumer is a widget that listens for changes in a class that implements `ChangeNotifier`, then rebuilds the widget tree below itself when it finds any.
- Whenever `notifyListeners()` is called, the Consumer's builder function is called. Synthax:

```
return
Consumer<ClassImplementingChangeNotifier>(
  builder: (context, cart, child) {
    return Text("Total price:
      ${cart.totalPrice}");
  },
);
```



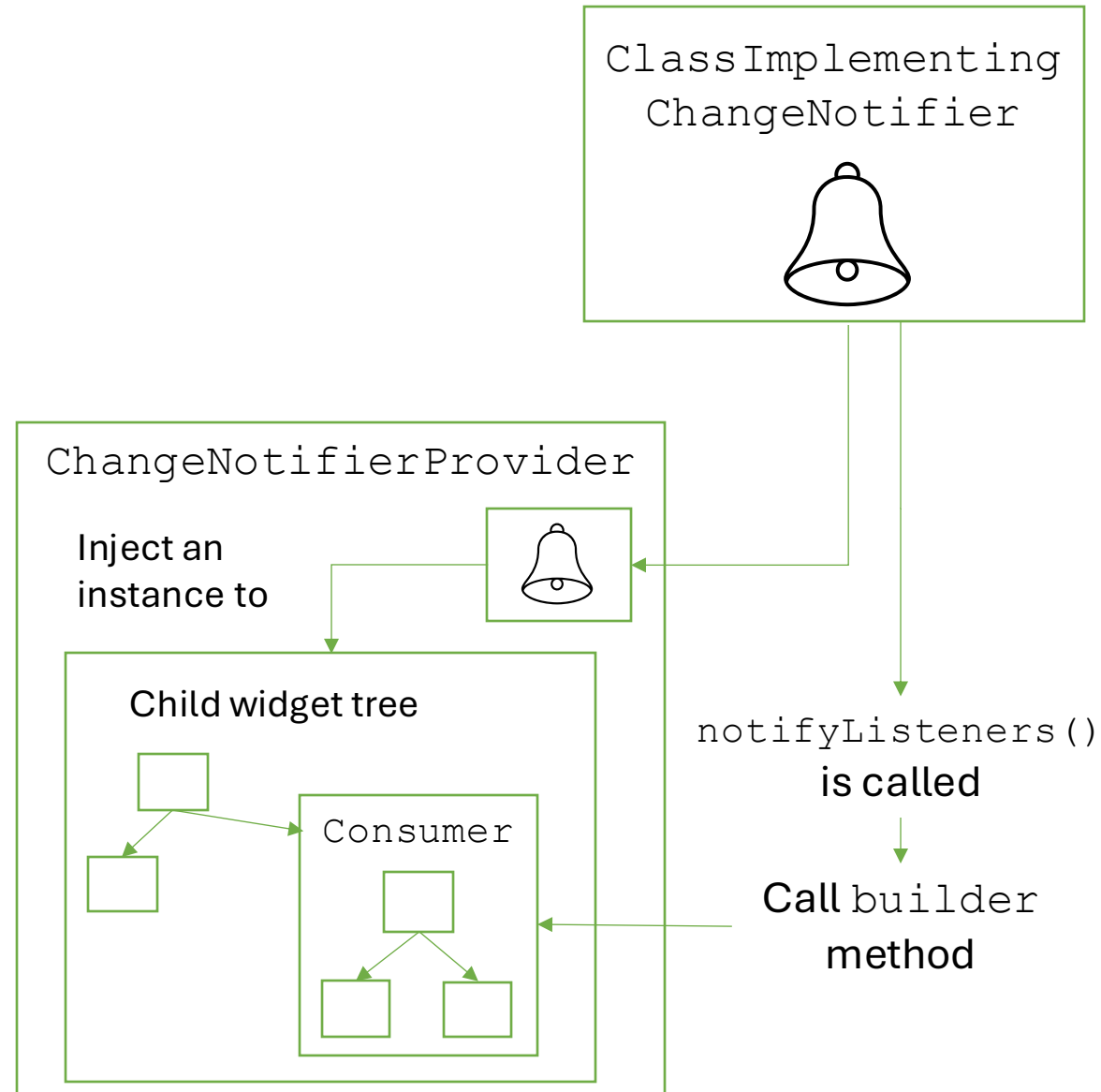
Provider classes: MultiProvider

- What if you need to inject more than one provider through the widget tree? Use `MultiProvider`.
- Synthax example:

```
MultiProvider (  
  providers: [  
    Provider<ChangeNotifier1>(create: (_) => ChangeNotifier1()),  
    Provider<ChangeNotifier2>(create: (_) => ChangeNotifier2()),  
    ...  
  ],  
  child: <widget_tree>,  
);
```

← List of providers.

Wrapping up: Provider (core) flow

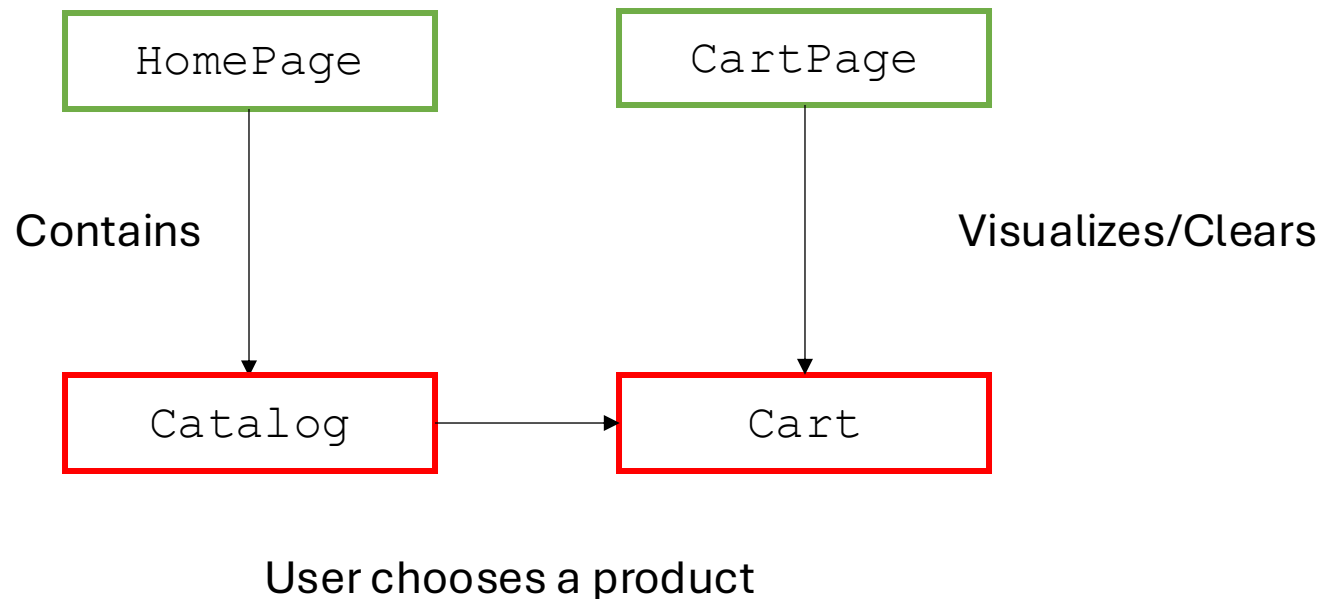


Outline

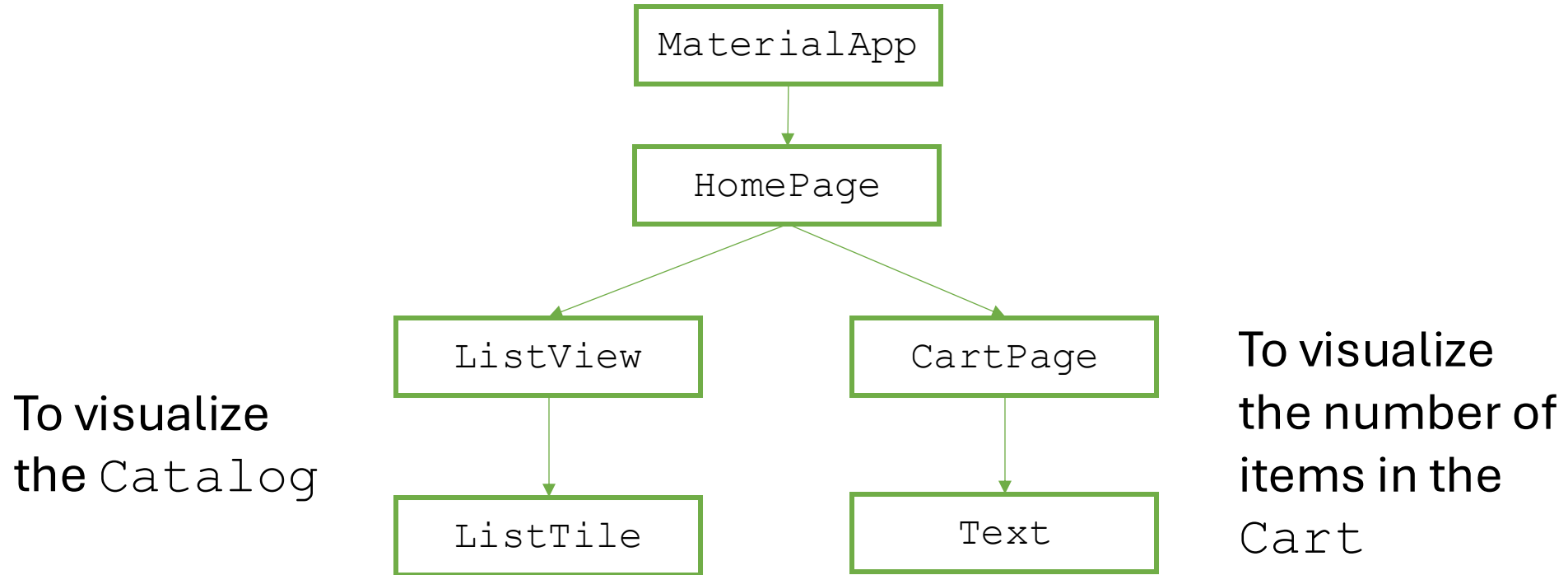
- State management concepts
- Provider
- **Case study**
- Exercise
- Homework
- Resources

Case of study

- We will build a (too) simple e-commerce app. Users can choose, in the `HomePage`, products from a catalog and add them to the cart. In a different screen, `CartPage`, the number of items in the cart is visualized.



Case of study – UI widget tree



➤ How to manage the catalog and the cart? With two class (model).

- No problems with the `Catalog` (Ephemeral state)...
- On the other hand, `Cart` must be shared (App state) Where to put it?

`Catalog`

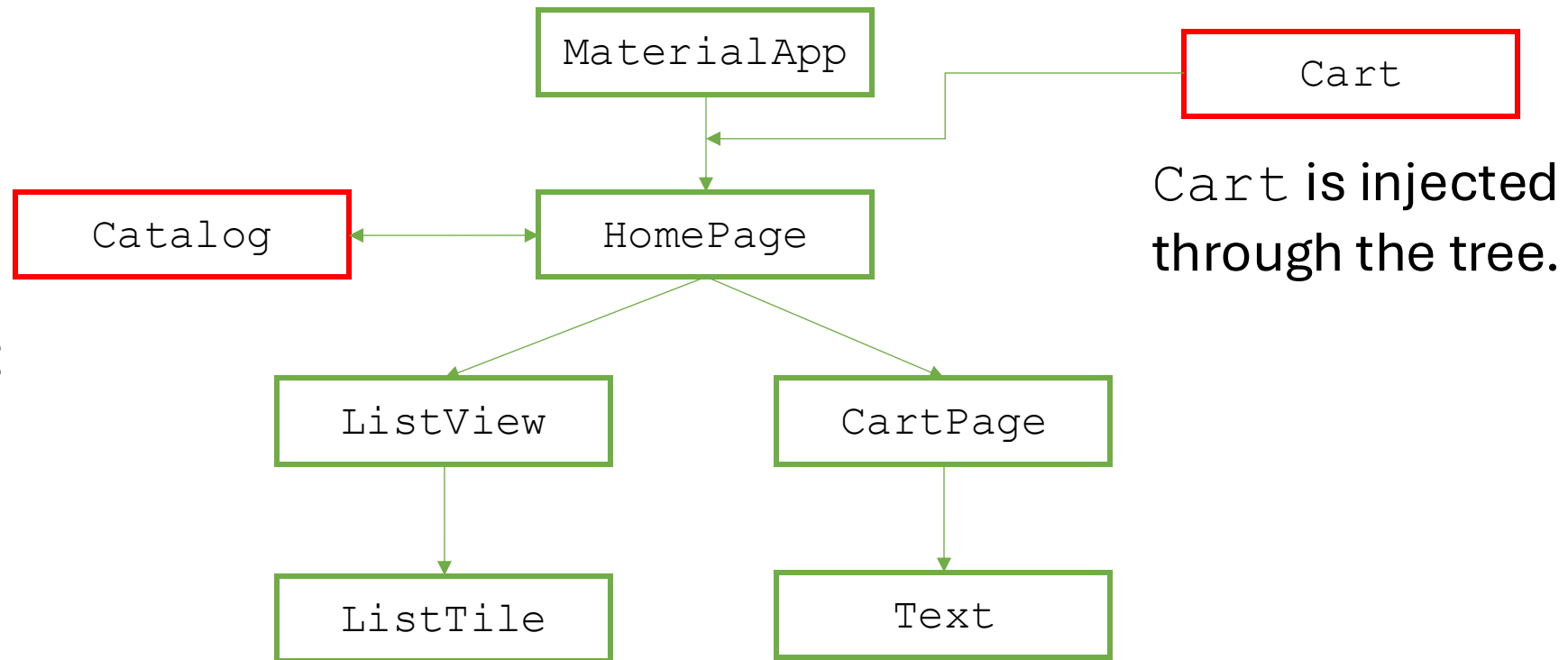
`Cart`

Lift the state up

- The idea is “Lift the state up”

$$\text{UI} = f(\text{state})$$

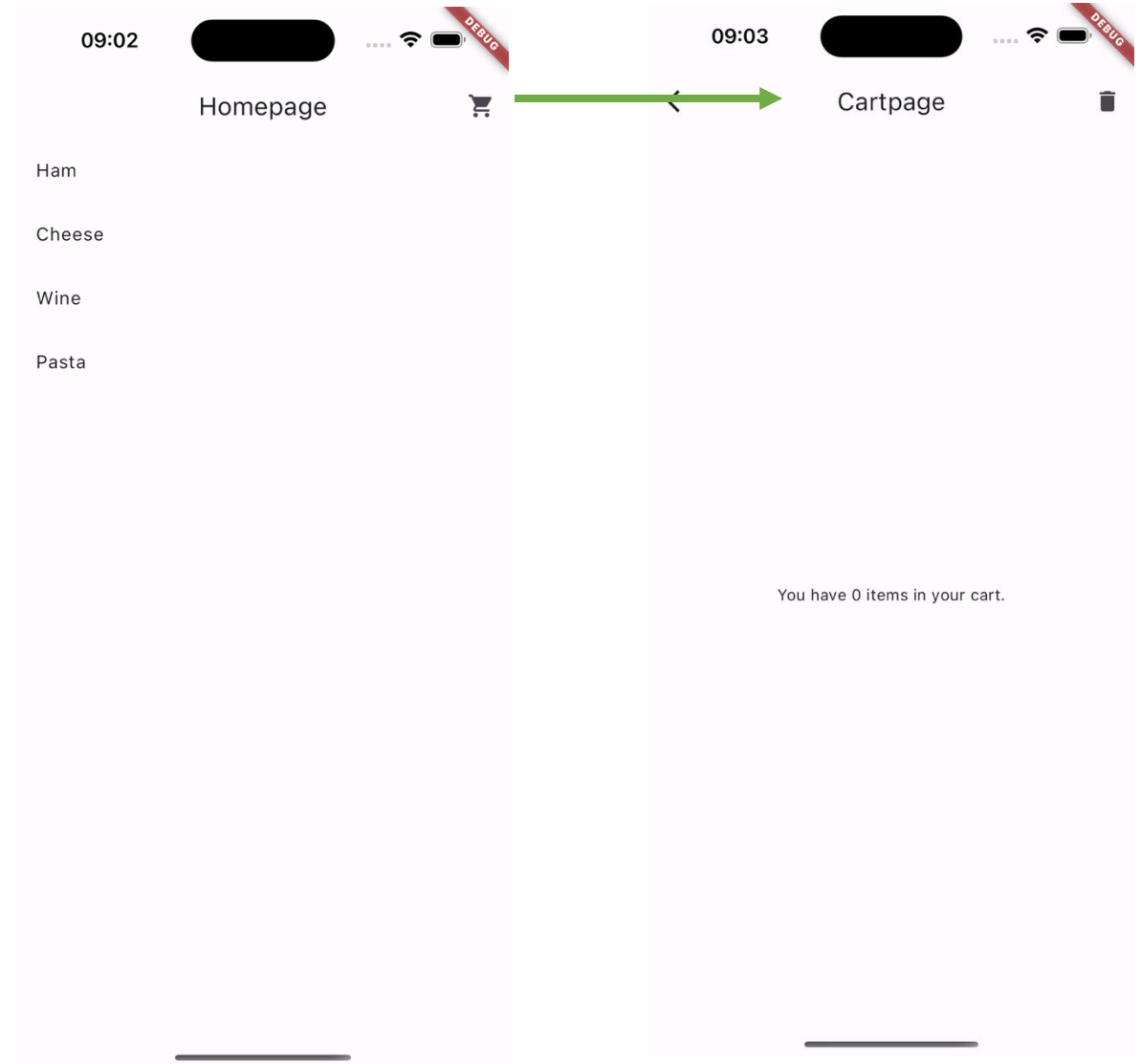
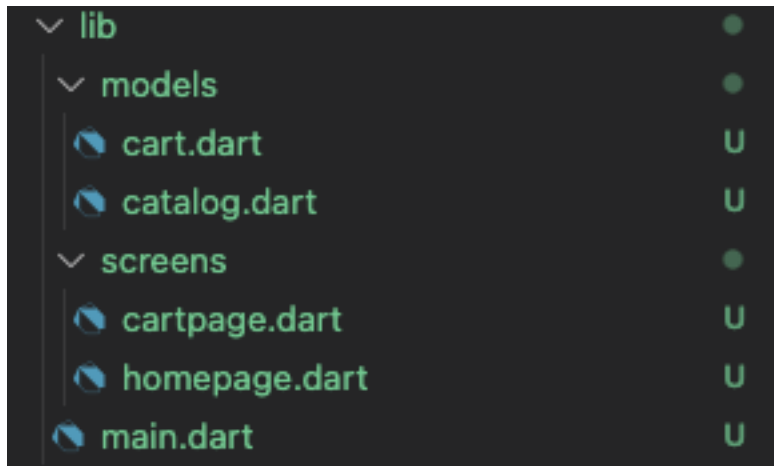
The layout on the screen Your build methods The application state



Catalog is just a variable of HomePage.

Case of study - Live

- Let's create a new project “shopper” that implements the case of study and shows how to use Provider.



Outline

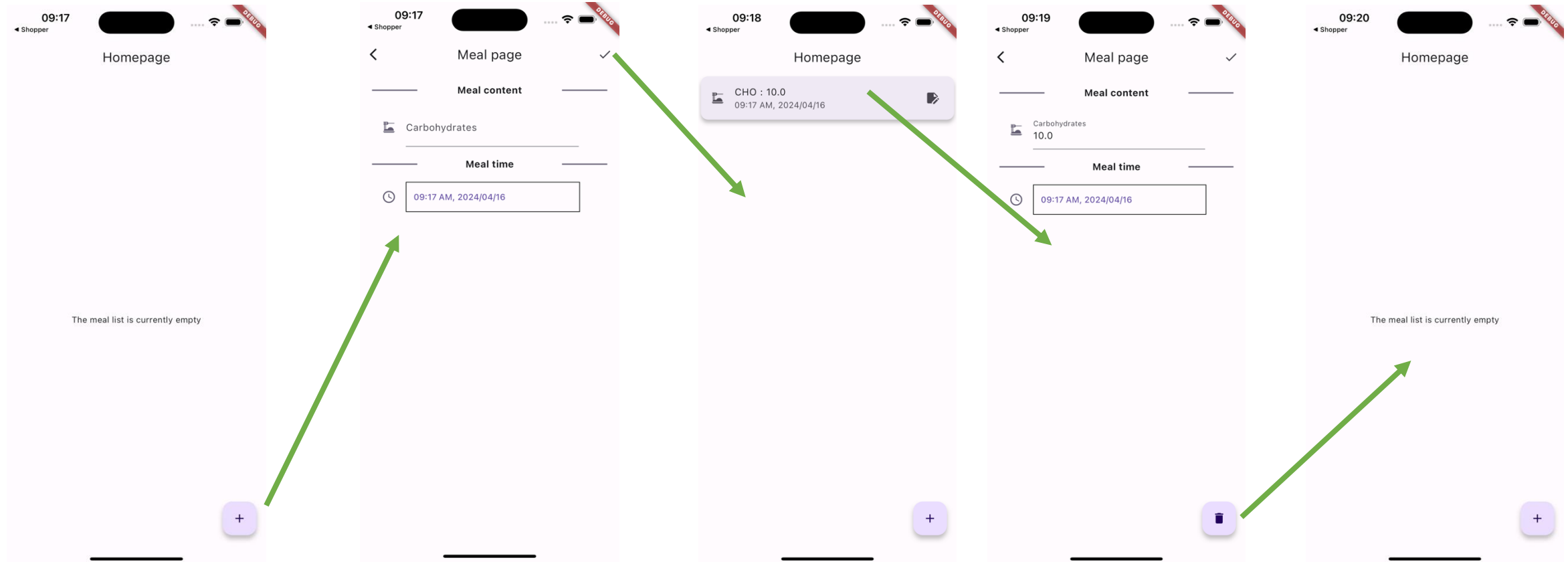
- State management concepts
- Provider
- Case study
- **Exercise**
- Homework
- Resources

Exercise

➤ Exercise 07.01

- Create a new project 'meal_manager', an app that stores the meal intakes of a user in terms of carbohydrate content and meal timing.
- The app needs to implement the following functionalities:
 - When a user opens (or restarts) the app, an empty list is showed;
 - By tapping a button, the user navigates to another page where he can add a new meal (CHO content and timing). Once he/she is done, the user taps a button and navigates back to the home page. The home page must show the updated meal list with the new meal just created;
 - The user can select a meal from the list. If he/she does that, he/she navigates to another page where he/she can edit or delete the meal entry from the list.

Exercise: here's a possible idea



Homepage is initially empty

When the button is tapped the user navigates to another page where he/she can add a new meal

When the user finishes, the new meal is added to the list

When the meal is tapped, the user navigates to a new screen where he/she can edit or delete it

If the meal is edited/deleted the list is updated.

Outline

- Recap
- State management concepts
- Provider
- Case study
- Exercise
- **Homework**
- Resources

Homework

- Get familiar with Provider
- Take a look to other approaches, e.g., Riverpod or BLoC. Maybe there is something that fits better your way of thinking!
- Take a look at my solution. You will find some useful code.

Outline

- State management concepts
- Provider
- Case study

- Exercise
- Homework
- **Resources**

Resources

- State management
 - <https://docs.flutter.dev/development/data-and-backend/state-mgmt/intro>
- Making sense of all those Flutter Providers
 - <https://medium.com/flutter-community/making-sense-all-of-those-flutter-providers-e842e18f45dd>
- List of state management approaches
 - <https://docs.flutter.dev/development/data-and-backend/state-mgmt/options>